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VLSM Subnetting Problem

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VLSM Subnetting Problem: 01

Question:

Suppose that we have three networks: A, B and C with IP requirements 50, 4 and 28. If you are given an IP block **130.3.0.0**, allocate IPs by performing VLSM subnetting.

Solution:

First arrange the networks according to descending order of IP requirements.

$$A(50), C(28), B(4)$$

For each subnet:

$$2^h \geq \text{Required IP}$$

where,

$$h = \text{Number of host bits}$$

and,

$$\text{Network Bits} = 32 - h$$

Subnet	No. of IPs Required	Host Bit Calculation	Allocated IPs	Host Bits / Net Bits	Subnet Mask	Allocated IP Range
A	50	$2^6 > 50 > 2^5$	64	$H = 6, N = 26$	255.255.255.192	130.3.0.0 – 130.3.0.63/26
C	28	$2^5 > 28 > 2^4$	32	$H = 5, N = 27$	255.255.255.224	130.3.0.64 – 130.3.0.95/27
B	4	$2^2 = 4$	4	$H = 2, N = 30$	255.255.255.252	130.3.0.96 – 130.3.0.99/30

Table 1: VLSM Allocation Table

Calculation Explanation

Network A

Required IP:

$$50$$

Find host bits:

$$2^6 = 64$$

Therefore,

$$h = 6$$

Network bits:

$$32 - 6 = 26$$

CIDR:

$$/26$$

Subnet Mask:

$$255.255.255.192$$

Network C

Required IP:

$$28$$

Host bits:

$$2^5 = 32$$

Network bits:

$$32 - 5 = 27$$

CIDR:

$$/27$$

Subnet Mask:

$$255.255.255.224$$

Network B

Required IP:

$$4$$

Host bits:

$$2^2 = 4$$

Network bits:

$$32 - 2 = 30$$

CIDR:

$$/30$$

Subnet Mask:

$$2$$

255.255.255.252

Total IP Utilization

Total Required IP:

$$50 + 28 + 4 = 82$$

Total Allocated IP:

$$64 + 32 + 4 = 100$$

Unused IP percentage:

$$\begin{aligned} &= \frac{100 - 82}{100} \times 100 \\ &= 18\% \end{aligned}$$

Therefore, using VLSM only **18% IP addresses are wasted**.

VLSM Subnetting Problem: 02

Question:

Suppose that we have four networks: A, B, C and D with IP requirements 120, 55, 25 and 10 respectively. If you are given an IP block **192.168.20.0**, allocate IPs by performing VLSM subnetting.

Solution:

Arrange the networks in descending order according to IP requirements.

A(120), B(55), C(25), D(10)

The required condition for host bits:

$$2^h \geq \text{Required IP Address}$$

where,

h = Number of Host Bits

Network bits are calculated as:

$$\text{Network Bits} = 32 - h$$

Subnet	No. of IPs Required	How many bits to borrow	No. of Allocated IPs	No. of Host Bits / Net Bits	Subnet Mask	Allocated IP Range
A	120	$2^7 > 120 > 2^6$	128	$H = 7, N = 25$	255.255.255.128	192.168.20.0 – 192.168.20.127/25
B	55	$2^6 > 55 > 2^5$	64	$H = 6, N = 26$	255.255.255.192	192.168.20.128 – 192.168.20.191/26
C	25	$2^5 > 25 > 2^4$	32	$H = 5, N = 27$	255.255.255.224	192.168.20.192 – 192.168.20.223/27
D	10	$2^4 > 10 > 2^3$	16	$H = 4, N = 28$	255.255.255.240	192.168.20.224 – 192.168.20.239/28

Table 2: VLSM Address Allocation

Detailed Calculation

Subnet A

Required IP:

$$120$$

Find minimum host bits:

$$2^7 = 128$$

Therefore,

$$h = 7$$

Network bits:

$$32 - 7 = 25$$

CIDR:

$$/25$$

Subnet Mask:

$$255.255.255.128$$

Subnet B

Required IP:

$$55$$

Host bits:

$$2^6 = 64$$

Network bits:

$$32 - 6 = 26$$

CIDR:

$$/26$$

Subnet Mask:

$$255.255.255.192$$

Subnet C

Required IP:

$$25$$

Host bits:

$$2^5 = 32$$

Network bits:

$$32 - 5 = 27$$

CIDR:

$$/27$$

Subnet Mask:

$$255.255.255.224$$

Subnet D

Required IP:

$$10$$

Host bits:

$$2^4 = 16$$

Network bits:

$$32 - 4 = 28$$

CIDR:

$$/28$$

Subnet Mask:

$$255.255.255.240$$

IP Utilization Calculation

Total required IP:

$$120 + 55 + 25 + 10 = 210$$

Total allocated IP:

$$128 + 64 + 32 + 16 = 240$$

Unused IP:

$$240 - 210 = 30$$

Percentage of unused IP:

$$= \frac{30}{240} \times 100$$

$$= 12.5\%$$

Therefore, after applying VLSM only **12.5% IP addresses are unused**.

VLSM Subnetting Problem: 03

Question:

Suppose that we have five networks: A, B, C, D and E with IP requirements 200, 90, 45, 18 and 6 respectively. If you are given an IP block **10.5.0.0**, allocate IP addresses by performing VLSM subnetting.

Solution

First arrange all networks according to descending order of IP requirements.

$$A(200), B(90), C(45), D(18), E(6)$$

For calculating host bits:

$$2^h \geq \text{Required IP}$$

where,

$$h = \text{Number of Host Bits}$$

Network bits:

$$\text{Network Bits} = 32 - h$$

Subnet	No. of IPs Required	How many bits to borrow	No. of Allocated IPs	No. of Host Bits / Net Bits	Subnet Mask	Allocated IP Range
A	200	$2^8 > 200 > 2^7$	256	$H = 8, N = 24$	255.255.255.0	10.5.0.0 – 10.5.0.255/24
B	90	$2^7 > 90 > 2^6$	128	$H = 7, N = 25$	255.255.255.128	10.5.1.0 – 10.5.1.127/25
C	45	$2^6 > 45 > 2^5$	64	$H = 6, N = 26$	255.255.255.192	10.5.1.128 – 10.5.1.191/26
D	18	$2^5 > 18 > 2^4$	32	$H = 5, N = 27$	255.255.255.224	10.5.1.192 – 10.5.1.223/27
E	6	$2^3 > 6 > 2^2$	8	$H = 3, N = 29$	255.255.255.248	10.5.1.224 – 10.5.1.231/29

Table 3: VLSM Subnet Allocation Table

Subnet Calculation

Subnet A

Required IP:

$$200$$

Host bits:

$$2^8 = 256$$

Therefore,

$$h = 8$$

Network bits:

$$32 - 8 = 24$$

CIDR:

$$/24$$

Subnet Mask:

$$255.255.255.0$$

Subnet B

Required IP:

$$90$$

Host bits:

$$2^7 = 128$$

Network bits:

$$32 - 7 = 25$$

CIDR:

$$/25$$

Subnet Mask:

$$255.255.255.128$$

Subnet C

Required IP:

$$45$$

Host bits:

$$2^6 = 64$$

Network bits:

$$32 - 6 = 26$$

CIDR:

/26

Subnet Mask:

255.255.255.192

Subnet D

Required IP:

18

Host bits:

$$2^5 = 32$$

Network bits:

$$32 - 5 = 27$$

CIDR:

/27

Subnet Mask:

255.255.255.224

Subnet E

Required IP:

6

Host bits:

$$2^3 = 8$$

Network bits:

$$32 - 3 = 29$$

CIDR:

/29

Subnet Mask:

255.255.255.248

IP Utilization

Total required IP:

$$200 + 90 + 45 + 18 + 6 = 359$$

Total allocated IP:

$$256 + 128 + 64 + 32 + 8 = 488$$

Unused IP:

$$488 - 359 = 129$$

Unused IP Percentage:

$$= \frac{129}{488} \times 100$$

$$= 26.43\%$$

Therefore, using VLSM only **26.43% IP addresses remain unused**.

VLSM Subnetting Problem: 04

Question:

Suppose that we have six networks: A, B, C, D, E and F with IP requirements 500, 250, 100, 60, 20 and 8 respectively. If you are given an IP block **172.25.0.0/16**, allocate IP addresses by performing VLSM subnetting.

Solution

First arrange all networks in descending order according to IP requirements.

$$A(500), B(250), C(100), D(60), E(20), F(8)$$

The number of host bits can be calculated using:

$$2^h \geq \text{Required IP Address}$$

where,

$$h = \text{Number of Host Bits}$$

Network bits:

$$\text{Network Bits} = 32 - h$$

Detailed Mathematical Calculation

Subnet A

Required IP:

$$500$$

Host bits:

Subnet	No. of IPs Required	How many bits to borrow	No. of Allocated IPs	No. of Host Bits / Net Bits	Subnet Mask	Allocated IP Range
A	500	$2^9 > 500 > 2^8$	512	$H = 9, N = 23$	255.255.254.0	172.25.0.0 – 172.25.1.255/23
B	250	$2^8 > 250 > 2^7$	256	$H = 8, N = 24$	255.255.255.0	172.25.2.0 – 172.25.2.255/24
C	100	$2^7 > 100 > 2^6$	128	$H = 7, N = 25$	255.255.255.128	172.25.3.0 – 172.25.3.127/25
D	60	$2^6 > 60 > 2^5$	64	$H = 6, N = 26$	255.255.255.192	172.25.3.128 – 172.25.3.191/26
E	20	$2^5 > 20 > 2^4$	32	$H = 5, N = 27$	255.255.255.224	172.25.3.192 – 172.25.3.223/27
F	8	$2^4 > 8 > 2^3$	16	$H = 4, N = 28$	255.255.255.240	172.25.3.224 – 172.25.3.239/28

Table 4: VLSM Allocation Table

$$2^9 = 512$$

Therefore,

$$h = 9$$

Network bits:

$$32 - 9 = 23$$

CIDR:

$$/23$$

Subnet Mask:

$$255.255.254.0$$

Subnet B

Required IP:

$$250$$

Host bits:

$$2^8 = 256$$

Network bits:

$$32 - 8 = 24$$

CIDR:

$$/24$$

Subnet Mask:

$$255.255.255.0$$

Subnet C

Required IP:

100

Host bits:

$$2^7 = 128$$

Network bits:

$$32 - 7 = 25$$

CIDR:

$$/25$$

Subnet Mask:

255.255.255.128

Subnet D

Required IP:

60

Host bits:

$$2^6 = 64$$

Network bits:

$$32 - 6 = 26$$

CIDR:

$$/26$$

Subnet Mask:

255.255.255.192

Subnet E

Required IP:

20

Host bits:

$$2^5 = 32$$

Network bits:

$$32 - 5 = 27$$

CIDR:

/27

Subnet Mask:

255.255.255.224

Subnet F

Required IP:

8

Host bits:

$$2^4 = 16$$

Network bits:

$$32 - 4 = 28$$

CIDR:

/28

Subnet Mask:

255.255.255.240

IP Utilization

Total Required IP:

$$500 + 250 + 100 + 60 + 20 + 8 = 938$$

Total Allocated IP:

$$512 + 256 + 128 + 64 + 32 + 16 = 1008$$

Unused IP:

$$1008 - 938 = 70$$

Percentage of unused IP:

$$= \frac{70}{1008} \times 100$$

$$= 6.94\%$$

Therefore, using VLSM only

6.94%

IP addresses remain unused.

VLSM Subnetting Problem: 05

Question:

Suppose that we have seven networks: A, B, C, D, E, F and G with IP requirements 750, 350, 180, 75, 35, 14 and 5 respectively. If you are given an IP block **192.100.0.0/16**, allocate IP addresses by performing VLSM subnetting.

Solution

Arrange the networks according to descending order of IP requirements:

$$A(750), B(350), C(180), D(75), E(35), F(14), G(5)$$

The required host bits are calculated using:

$$2^h \geq \text{Required IP Address}$$

where,

$$h = \text{Number of Host Bits}$$

Network bits:

$$\text{Network Bits} = 32 - h$$

Subnet	No. of IPs Required	How many bits to borrow	Allocated IPs	Host Bits / Net Bits	Subnet Mask	Allocated IP Range
A	750	$2^{10} > 750 > 2^9$	1024	$H = 10, N = 22$	255.255.252.0	192.100.0.0–192.100.3.255/22
B	350	$2^9 > 350 > 2^8$	512	$H = 9, N = 23$	255.255.254.0	192.100.4.0–192.100.5.255/23
C	180	$2^8 > 180 > 2^7$	256	$H = 8, N = 24$	255.255.255.0	192.100.6.0–192.100.6.255/24
D	75	$2^7 > 75 > 2^6$	128	$H = 7, N = 25$	255.255.255.128	192.100.7.0–192.100.7.127/25
E	35	$2^6 > 35 > 2^5$	64	$H = 6, N = 26$	255.255.255.192	192.100.7.128–192.100.7.191/26
F	14	$2^4 > 14 > 2^3$	16	$H = 4, N = 28$	255.255.255.240	192.100.7.192–192.100.7.207/28
G	5	$2^3 > 5 > 2^2$	8	$H = 3, N = 29$	255.255.255.248	192.100.7.208–192.100.7.215/29

Table 5: VLSM Subnet Allocation Table

Mathematical Calculation

Subnet A

Required IP:

$$750$$

Host bits:

$$2^{10} = 1024$$

Network bits:

$$32 - 10 = 22$$

CIDR:

/22

Subnet B

Required IP:

350

Host bits:

$$2^9 = 512$$

Network bits:

$$32 - 9 = 23$$

CIDR:

/23

Subnet C

Required IP:

180

Host bits:

$$2^8 = 256$$

Network bits:

$$32 - 8 = 24$$

CIDR:

/24

Subnet D

Required IP:

75

Host bits:

$$2^7 = 128$$

Network bits:

$$32 - 7 = 25$$

CIDR:

/25

Subnet E

Required IP:

35

Host bits:

$$2^6 = 64$$

Network bits:

$$32 - 6 = 26$$

CIDR:

/26

Subnet F

Required IP:

14

Host bits:

$$2^4 = 16$$

Network bits:

$$32 - 4 = 28$$

CIDR:

/28

Subnet G

Required IP:

5

Host bits:

$$2^3 = 8$$

Network bits:

$$32 - 3 = 29$$

CIDR:

/29

IP Utilization Calculation

Total Required IP:

$$750 + 350 + 180 + 75 + 35 + 14 + 5 = 1409$$

Total Allocated IP:

$$1024 + 512 + 256 + 128 + 64 + 16 + 8 = 2008$$

Unused IP:

$$2008 - 1409 = 599$$

Unused IP Percentage:

$$= \frac{599}{2008} \times 100$$

$$= 29.83\%$$

Therefore, after applying VLSM:

29.83%

IP addresses remain unused.